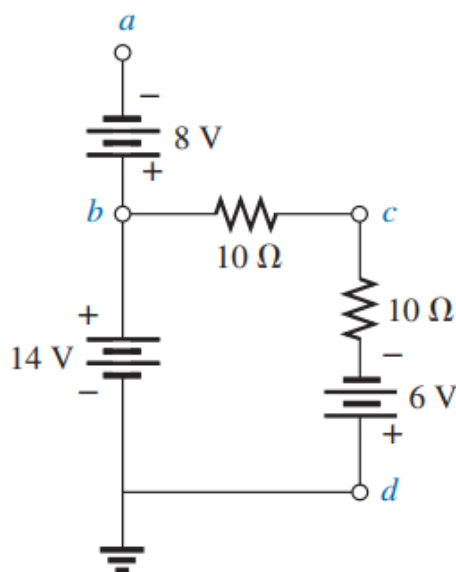


CSE250: Circuits and Electronics

Practice Problems Set 2

1. For the network shown below, determine,

- a. V_a, V_b, V_c, V_d
- b. V_{ab}, V_{cb}, V_{cd}
- c. V_{ad}, V_{ca}

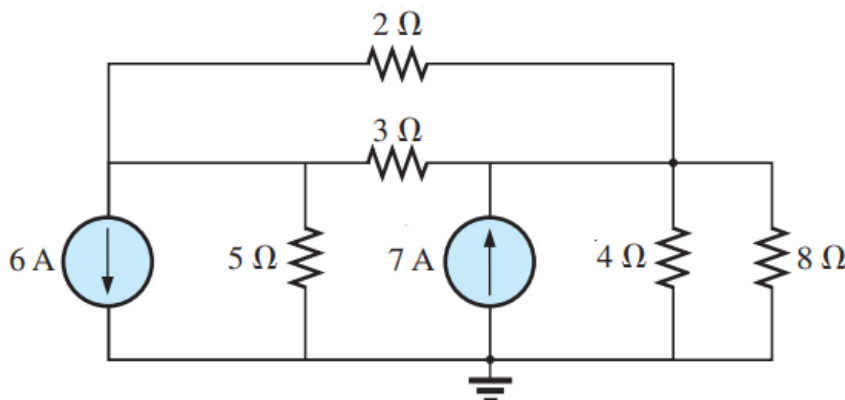


Answer:

- a. 6 V, 14 V, 4 V, 0 V
- b. -8 V, -10 V, 4 V
- c. 6 V, -2 V

2. For the network shown below,

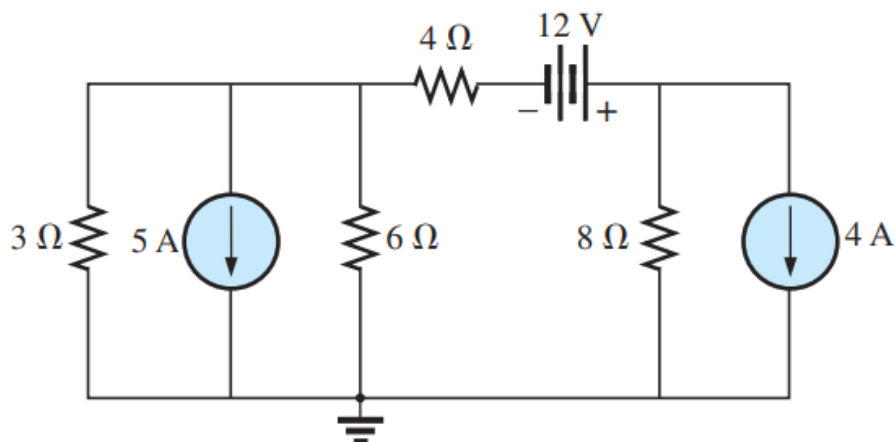
- a. Count the number of nodes.
- b. Determine the node voltages.
- c. Determine the power supplied by the current source.



Answer:

- a. Try yourself.
- b. -2.56 V, 4.03 V, 0 V

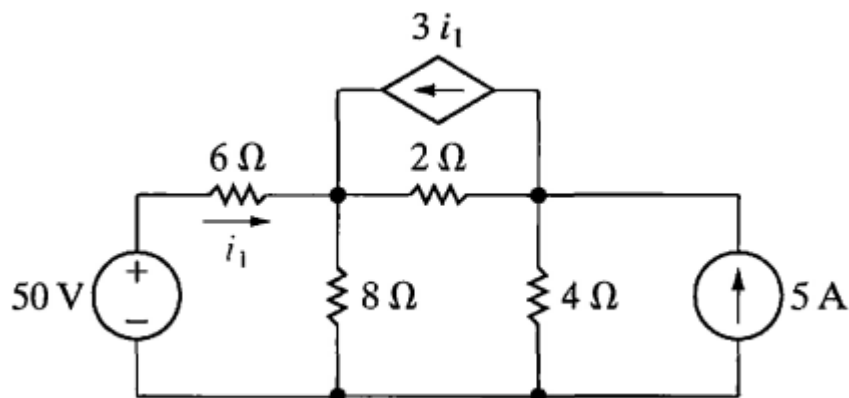
3. Use nodal analysis and determine the power of the 12 V battery. **Answer:** -29.14 W



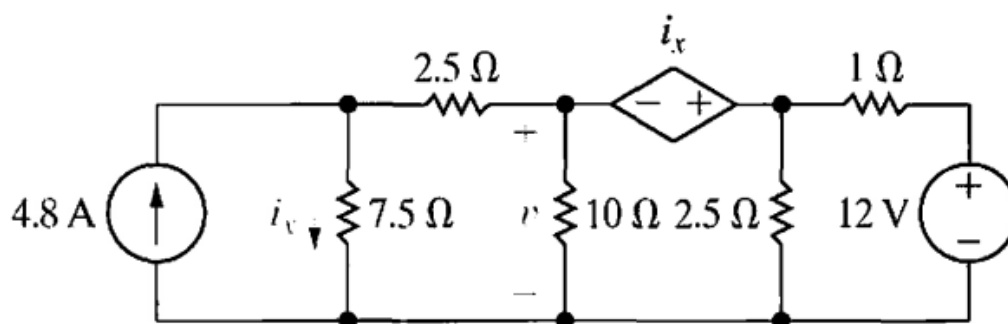
4. Use the node-voltage method to find the power associated with each source in the circuit shown.

Answer:

$$p_{50V} = -150 \text{ W}, p_{3i_1} = -144 \text{ W}, p_{5A} = -80 \text{ W};$$

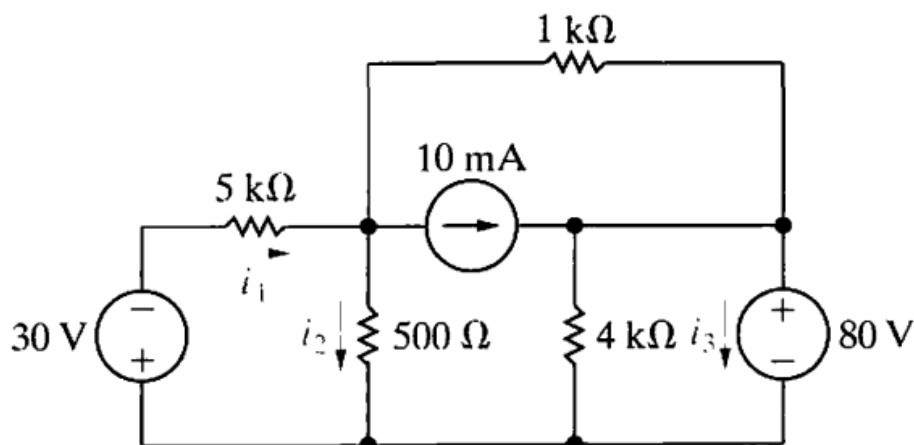


5. Use nodal analysis and find the current through the dependent voltage source. **Answer:** $\pm 2 \text{ A}$



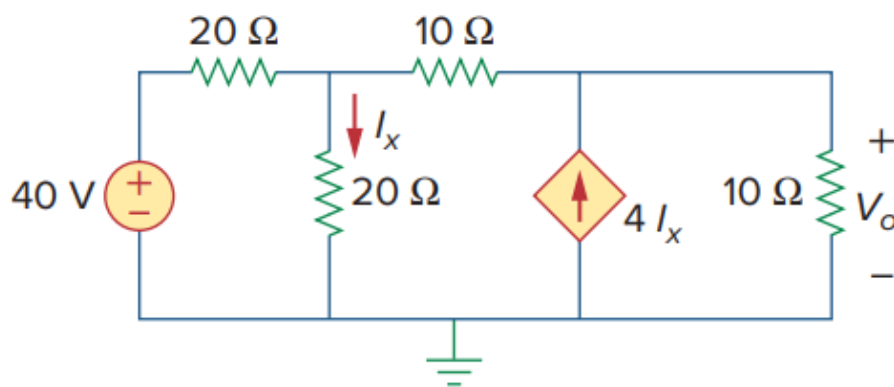
6. Use the node-voltage method to find the branch currents i_1 , i_2 , and i_3 .

Answer: 2 mA, 40 mA, -70 mA



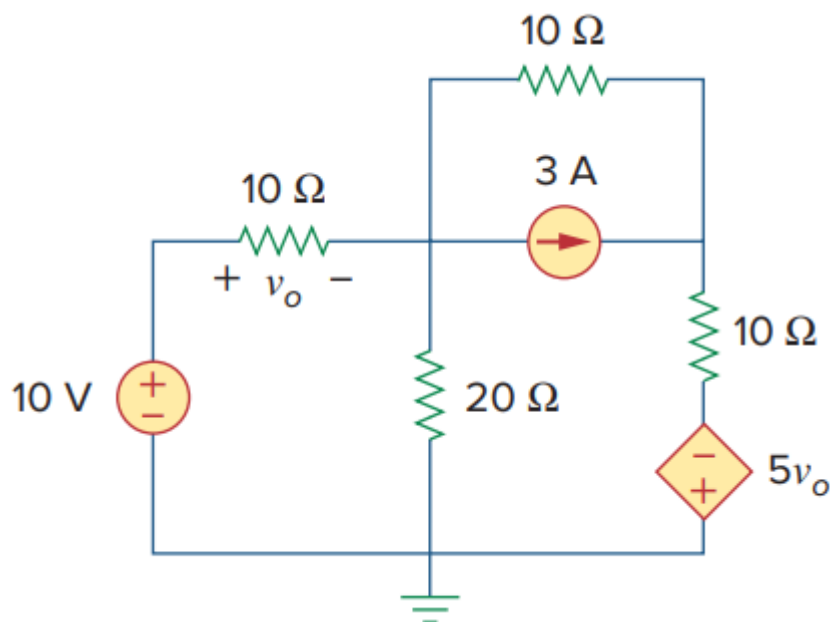
7. Using nodal analysis, determine V_o in the circuit below.

Answer: 60 V



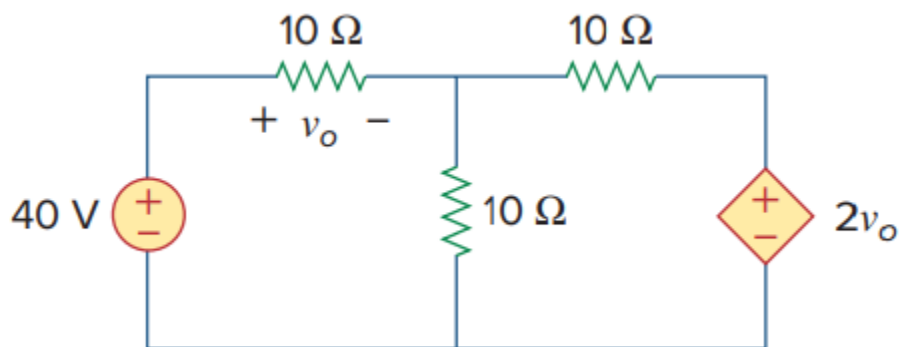
8. Using nodal analysis, determine v_o in the circuit below.

Answer: -50 V



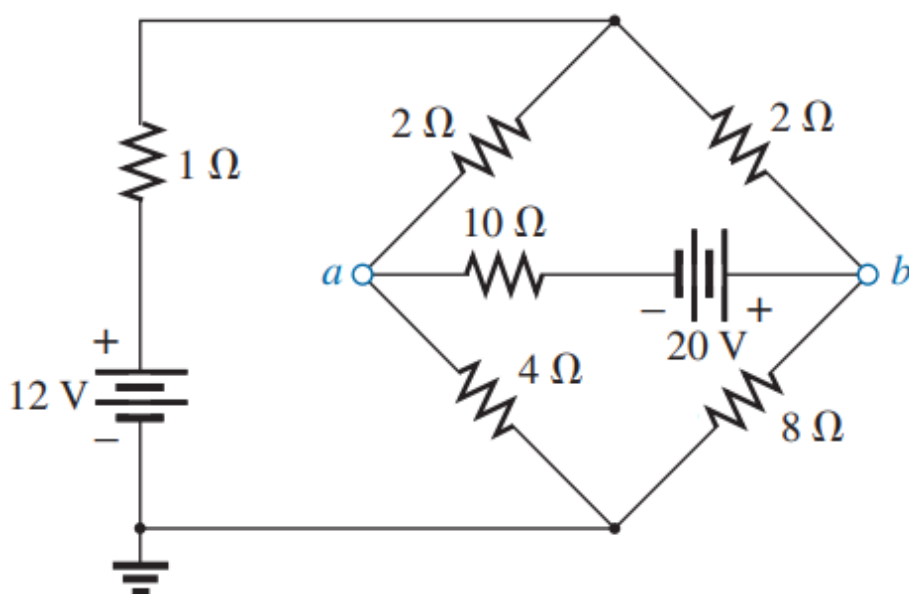
9. Using mesh analysis, find v_o .

Answer: 16 V



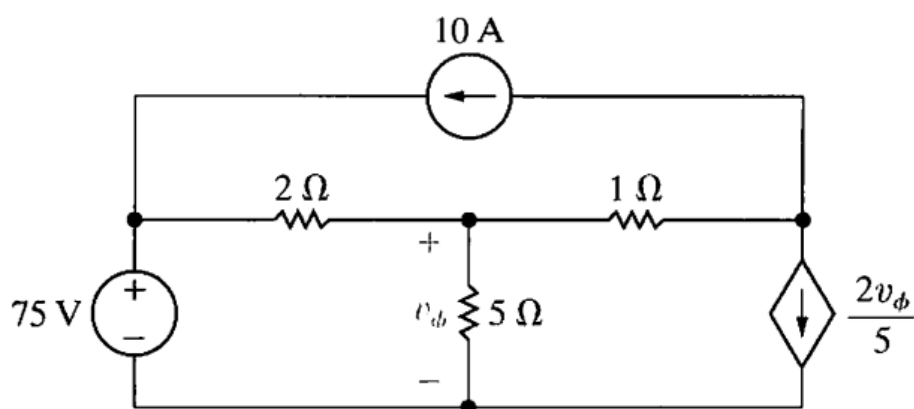
10. Use mesh analysis and determine V_{ab} .

Answer: -5.53 V

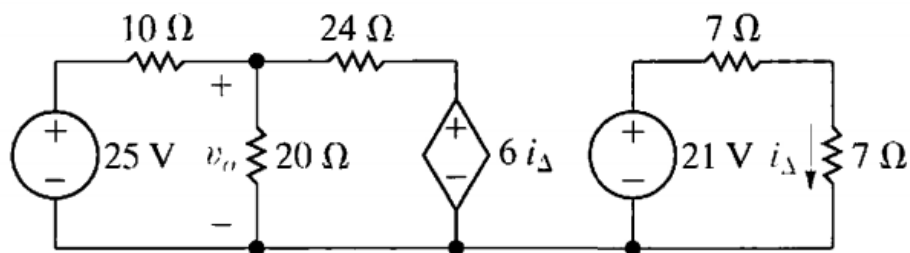


11. Use mesh analysis to determine the current supplied by the 75 V source.

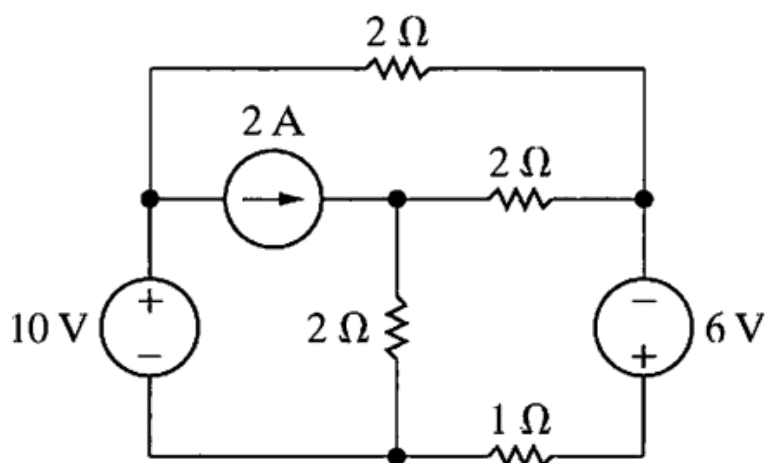
Answer: 15 A



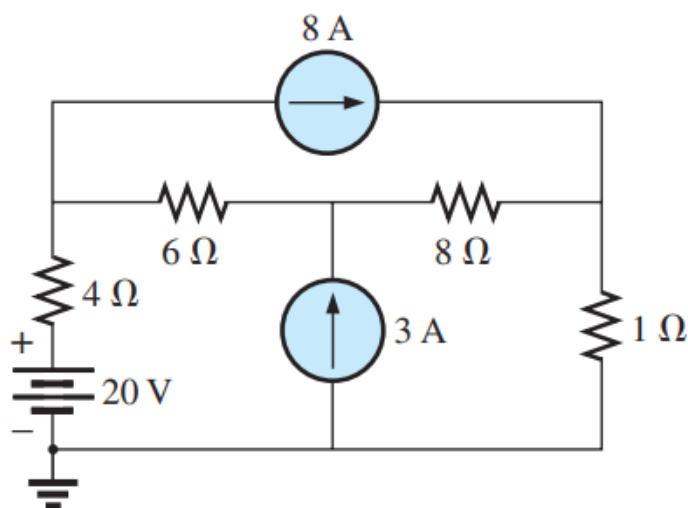
- 12.** Use mesh analysis and determine v_o . Find the power delivered by the dependent source. **Answer:** 15 V



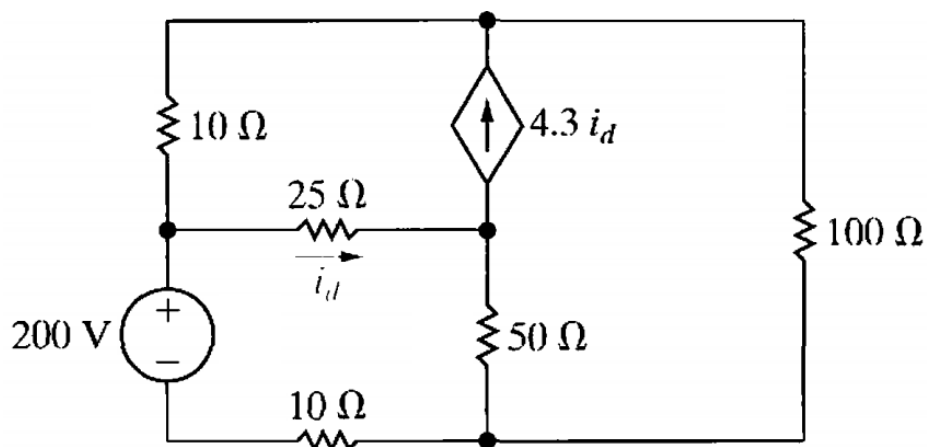
- 13.** Use the mesh-current method to find the power dissipated in the 1 Ω resistor in the circuit shown. **Answer:** 36 W.



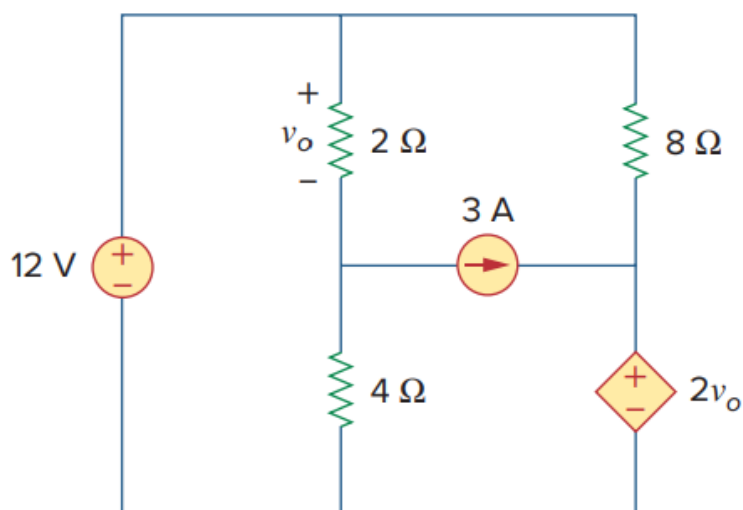
- 14.** Use mesh analysis and determine the power of the 3 A source. **Answer:** 12.74 W



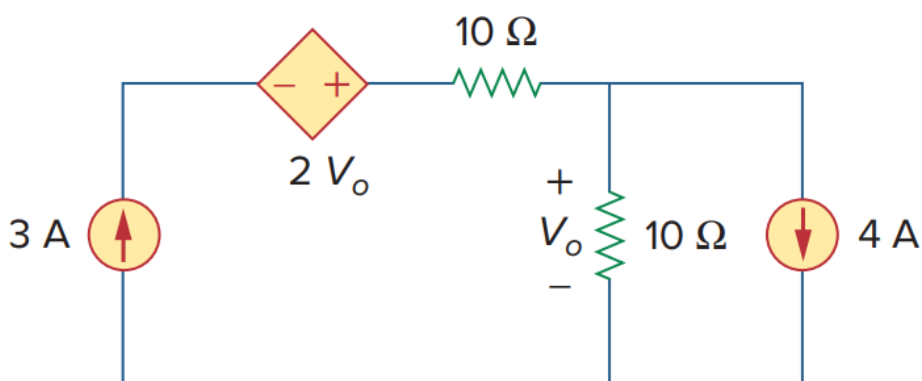
- 15.** Use mesh analysis and determine the power of the dependent source. **Answer:** -0.75 kW



- 16.** Use mesh analysis method and determine voltage across the 3 A current source. **Answer:** $\mp 12 \text{ V}$

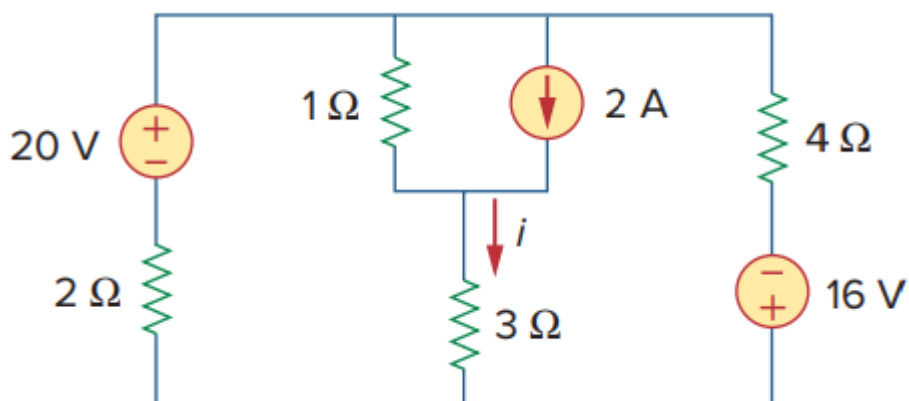


- 17.** Use superposition to solve for v_o in the following circuit. **Answer:**
 $v_o = -10 \text{ V}$



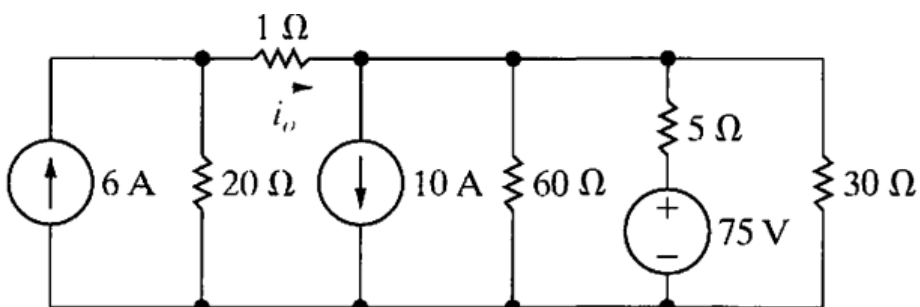
- 18.** For the circuit shown below, use superposition to find i . Calculate the power delivered to the 3- Ω resistor.

Answer:
1.875 A, 10.55 W



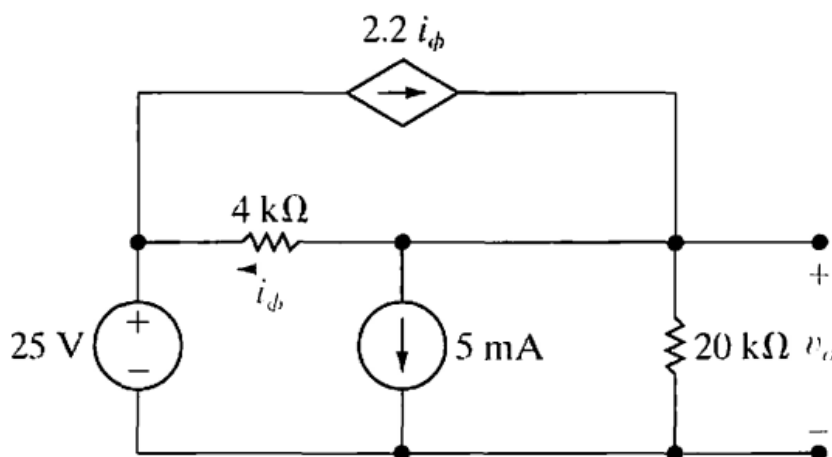
- 19.** Use the superposition principle to find i_0 in the following circuit.

Answer:
 $i_0 = 4.8 + 1.6 - 2.4$
 $= 4 \text{ A}$



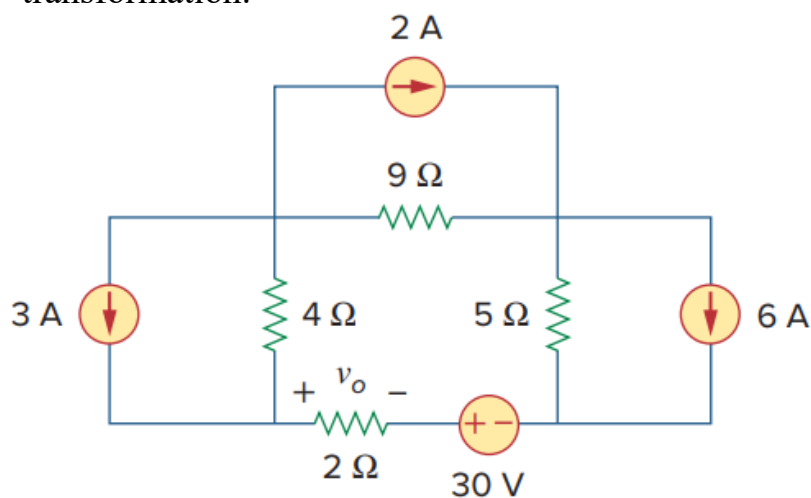
- 20.** Determine v_0 using superposition principle.

Answer:
 $v_0 = 30 + 20$
 $= 50 \text{ V}$



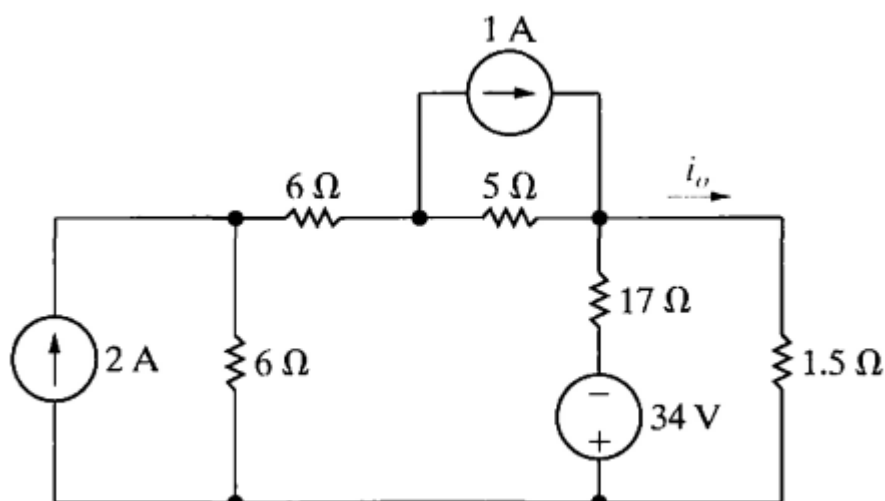
- 21.** Obtain v_o in the following circuit using source transformation.

Answer: -6.6 V



- 22.** Use a series of source transformations to find i_o .

Answer: -0.85 A



- 23.** Use source transformation to find v_o .

Answer: 3 V

